

LAB ACTIVITY: College Application Evaluator

Lab 2: College Application Evaluator

Overview

You'll write a single Python script that models how a college might screen an incoming application: minimum admission requirements, automatic admission, a tiered academic scholarship, Honors College eligibility, major-based placement, a leadership award, a first-generation support program, and in-state/out-of-state tuition handling.

Skills this lab is checking: `input()`, `print()`, output, `if/elif/else`, and comparison/boolean expressions.

Required Sections

Your script must contain **Milestones 1-5**, in order, on one file.

Milestone 1 (20 pts): Applicant Information

- 1) Prompt for:
 - Applicant **name** and **state of residence**.
 - **High school GPA** and **ACT score** (overall).
 - **ACT Math score** and **ACT English score**.
 - **Community service hours** and **number of extracurricular activities**.
 - Whether the applicant is a **first-generation college student** (*yes/no*).
 - **Intended major**.
- 2) Convert every numeric response to the correct type (`float` for GPA, `int` for everything else) at the moment you read it in.
- 3) Generate output to match the example below, including the Application Results for <name> header.

Milestone 2 (20 pts): Basic Requirements and Admission Decision

- 1) Check the **GPA requirement**: met if GPA is 2.5 or higher.
- 2) Check the **ACT requirement**: met if the ACT score is 18 or higher.
- 3) Determine the overall **admission status** — a student is ADMITTED only if *both* the GPA and ACT requirements are met.
- 4) Generate output to match the example below.

Milestone 3 (20 pts): Automatic Admission and Academic Scholarship

- 1) Determine **automatic admission**: YES only if GPA is 3.5 or higher *and* ACT is 24 or higher.
- 2) Determine the applicant's **academic scholarship** using the following tiers, checked from highest to lowest so that only the *best* tier a student qualifies for is awarded:

GPA at least	ACT at least	Scholarship
3.9	32	\$12,000
3.7	28	\$8,000
3.5	25	\$5,000
3.2	22	\$2,000
(none of the above)		\$0

Start a `scholarship` variable at 0, and only check a lower tier if `scholarship` is still 0 after checking every tier above it.

- 3) Generate output to match the example below.

Milestone 4 (20 pts): Placement and Recognition

- 1) Determine **Honors College** eligibility: ELIGIBLE only if GPA is 3.7 or higher *and* ACT is 28 or higher.
- 2) Determine **Engineering placement**, but only print anything for this section if the intended major (compared case-insensitively) is "engineering": READY FOR CALCULUS TRACK if the ACT Math score is 24 or higher, otherwise MATH PLACEMENT REQUIRED.
- 3) Determine **English placement** for every applicant based on ACT English score: ADVANCED COMPOSITION (26+), COMPOSITION I (20-25), or DEVELOPMENTAL SUPPORT RECOMMENDED (below 20).
- 4) Determine the **community leadership award**: ELIGIBLE only if community service hours are 100 or more *and* the applicant lists 3 or more extracurricular activities.
- 5) Determine **first-generation support program** eligibility based directly on the applicant's yes/no answer (compared case-insensitively).
- 6) Generate output to match the example below.

Milestone 5 (20 pts): Tuition and Presidential Scholarship

- 1) Determine **tuition classification**: IN-STATE if the applicant's state (compared case-insensitively) is your home state, otherwise OUT-OF-STATE.
- 2) For **out-of-state** applicants only, determine **out-of-state tuition waiver** eligibility: ELIGIBLE only if GPA is 3.5 or higher *and* ACT is 26 or higher.
- 3) Determine eligibility for a **Presidential Scholarship interview**: only print an invitation line, and only if GPA is 3.9 or higher *and* ACT is 30 or higher *and* community service hours are 50 or more. If the applicant doesn't qualify, print nothing for this section.
- 4) Generate output to match the example below.

Deliverables

Submit all of the following together:

- 1) **One .py file** containing all five sections, including comments and blank lines for spacing.
- 2) Your code should be commented. Not commenting your code will result in -10 points. Comment headers should mark each numbered requirement (matching Milestones 2-5 above) so a grader can find each one quickly.
- 3) To receive credit for the assignment, you need to complete a check off with one of the lab TAs.
 - The TA will ask you questions about your code. If you cannot answer the TA's questions, you will not receive credit for the assignment. Checkoffs can be attempted multiple times.

Example Output 1

```
Applicant name: Taylor Morgan
State of residence: Alabama
High school GPA: 3.95
ACT score: 32
ACT Math score: 30
ACT English score: 27
Community service hours: 120
Number of extracurricular activities: 4
First-generation college student? (yes/no): no
Intended major: Engineering
```

```
=====
College Application Evaluator
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```

```
Application Results for Taylor Morgan
-----
GPA requirement: MET
ACT requirement: MET
Admission status: ADMITTED
Automatic admission: YES
Academic scholarship: $12000
Honors College: ELIGIBLE
Engineering placement: READY FOR CALCULUS TRACK
English placement: ADVANCED COMPOSITION
Leadership award: ELIGIBLE
First-generation support program: NOT ELIGIBLE
Tuition classification: OUT-OF-STATE
```

Out-of-state tuition waiver: ELIGIBLE
Presidential Scholarship interview: INVITED

Application evaluation complete.

(The prompts above are shown before the banner only to illustrate what the program asks for — in your actual program, `print()` the banner first, then prompt for input, matching the order in Milestone 2.)

Example 1 is a strong out-of-state applicant, so it shows what happens when almost every branch's best-case line prints, including the Presidential Scholarship invitation and a full \$12,000 scholarship award.

Example Output 2

Applicant name: Priya Nair
State of residence: Mississippi
High school GPA: 3.6
ACT score: 25
ACT Math score: 21
ACT English score: 24
Community service hours: 80
Number of extracurricular activities: 2
First-generation college student? (yes/no): yes
Intended major: Biology

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Application Results for Priya Nair

GPA requirement: MET
ACT requirement: MET
Admission status: ADMITTED
Automatic admission: YES
Academic scholarship: \$5000
Honors College: NOT ELIGIBLE
English placement: COMPOSITION I
Leadership award: NOT ELIGIBLE
First-generation support program: ELIGIBLE
Tuition classification: IN-STATE

Application evaluation complete.

Example 2 is an in-state, non-Engineering applicant whose GPA and ACT land in the middle of the scholarship table. Notice two things this example demonstrates that Example 1 doesn't: the **Engineering placement** line is completely absent (the major isn't Engineering, so that section prints nothing), and there's no out-of-state tuition waiver or Presidential Scholarship line at all, because both of those sections only ever print for an out-of-state applicant.

Example Output 3

Applicant name: Devon Marsh
State of residence: Tennessee
High school GPA: 2.0
ACT score: 16

ACT Math score: 18
ACT English score: 17
Community service hours: 20
Number of extracurricular activities: 1
First-generation college student? (yes/no): no
Intended major: Undecided

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Application Results for Devon Marsh

GPA requirement: NOT MET
ACT requirement: NOT MET
Admission status: NOT ADMITTED
Automatic admission: NO
Academic scholarship: \$0
Honors College: NOT ELIGIBLE
English placement: DEVELOPMENTAL SUPPORT RECOMMENDED
Leadership award: NOT ELIGIBLE
First-generation support program: NOT ELIGIBLE
Tuition classification: OUT-OF-STATE
Out-of-state tuition waiver: NOT ELIGIBLE

Application evaluation complete.

Example 3 is the required “fails the basic requirements” case from the self-check below. It’s also out-of-state, so unlike Example 2, the out-of-state tuition waiver line *does* print here — just as NOT ELIGIBLE, since the GPA/ACT minimums for the waiver aren’t met either. This is the example to compare your own output against if a section you expect to see is missing or a section you expect to be silent is printing anyway.

self-check

- All 5 Milestones are present, in order, and clearly labeled.
- Every scholarship tier is checked from highest to lowest, and only one tier is ever awarded.
- The Engineering placement section prints nothing for a non-Engineering major, and the Presidential Scholarship line prints nothing when the applicant doesn’t qualify.
- You ran the script at least three times with different inputs — including once with an applicant who fails the basic GPA/ACT requirements — and verified the outputs against Examples 1-3 above.